

PDF rendering core & region registry

The rendering layer is the bridge between raw PDFs in the raw corpus folder and the structured registry of regions in the Promoted Regions Database. It has no LLM dependency — it exposes deterministic primitives that the two cropping agents call as tools.

Rendering primitives

Three functions cover every render the pipeline ever performs.

Primitive	Purpose	Output
Page info	page count, per-page dimensions (pt + mm), rotation, whether the page carries extracted text	JSON metadata
Grid overview	renders a full page at 120 DPI with a normalized coordinate grid overlay (alphanumeric cell labels starting at <code>A1</code> , axis labels <code>0.0</code> – <code>1.0</code> ; density adapts to page size)	saved to the grid directory; returned with an inline JPEG preview
High-resolution crop	renders a chosen bbox at 150–600 DPI as PNG; tight clamping and padding in normalized units	saved to the configured output directory; returned with an inline JPEG preview

Coordinate system

- All coordinates are normalized `[0, 1]` on the **rotated visible page** (origin top-left).
- The `units` parameter accepts `norm` / `pt` / `mm` ; padding is always applied in normalized units regardless of input.

Pixel cap

- A single render may not exceed **40 megapixels**.
- If a requested crop's dimensions exceed this 40 Mpx safety cap, the rendering engine **automatically and dynamically downscales the rendering DPI** on the fly to the maximum safe threshold. It renders and saves the crop successfully under this safe resolution, logging a clear warning notice inside the output metadata to maintain complete system auditability.

Scratch staging

- Agent renders land in a temporary scratch directory while the agent iterates on the bbox.
- Only the file the agent explicitly commits (via the draft region tool) is promoted to the canonical crop name.
- A wrong commit is overwritten atomically on the next commit; iteration drafts never pollute the canonical directory.

Corpus discovery

The corpus is the source-of-truth and is never modified by the pipeline.

- Any PDF in the raw corpus folder is in scope. No hardcoded file lists.
- `doc_id` is derived from the filename: uppercase, non-alphanumeric collapsed to underscore, capped at 64 characters.
- Path resolution accepts a `doc_id`, a repo-relative path, an absolute path, or a bare filename.
- The repo root is detected automatically; an environment variable override is available.

Error model

The rendering primitives surface a small set of named errors. The agent tool handlers catch all of them and serialize them into a `{ok: false, error: ...}` JSON tool result, so the agent can recover and try a different bbox.

Error	When
Path-not-in-corpus	The path doesn't resolve to any PDF under the raw corpus folder
Page-range error	The requested page number exceeds the document's page count
Bad-box error	Coordinate box is invalid (wrong shape, degenerate, inverted)
(Auto-scaled)	Large crops exceeding 40 Mpx are automatically downscaled on the fly rather than throwing an error

Region registry

Every approved crop is registered as one entry in the Promoted Regions Database.

Fields

Field	Purpose
<code>id</code>	deterministic ID: <code>{doc_id}_p{page:02d}_{slug}</code> ; collisions get <code>_2</code> , <code>_3</code> , ...
<code>doc_id</code> , <code>page</code>	source document and 1-indexed page
<code>bbox_norm</code>	normalized <code>[x1, y1, x2, y2]</code>
<code>label</code>	short human-readable name (also drives the URN slug)
<code>description</code>	one sentence written by the cropping agent at creation
<code>crop_file</code>	repo-relative path to the canonical PNG
<code>source_type</code>	format class + document title; copied onto the asset at URN minting
<code>caption</code> , <code>vision_prompt</code>	populated automatically after region approval

Field	Purpose
<code>needs_verification</code>	informational — crop quality not yet checked
<code>HITL_approval</code>	the only gate; only approved regions reach URN minting and indexing

Lifecycle

Stage	<code>needs_verification</code>	<code>HITL_approval</code>
Just proposed	<code>true</code>	<code>false</code>
After the verifier passes (or 9 rounds exhausted)	<code>false</code>	<code>false</code>
After human review at Gate 3	<code>false</code>	<code>true</code>

Atomic-save guarantee

Region writes always go through a **write-`.tmp`-then-rename** path. The in-memory cache is invalidated explicitly before every agent commit — sub-second back-to-back writes within the same mtime tick would otherwise silently lose one update.

Machine catalog export

The catalog export step is the **only egress point** from the library. It runs the integrity check internally first; on any failure the export refuses and exits non-zero. One catalog file is emitted per distinct machine in the inventory; the output directory is pre-cleaned of stale machine catalog files so a removed machine doesn't linger.

Schema (one file per machine)

```
{
  "machine_id": "3048",
```

```

"exported_at": "2026-05-05T14:00:00Z",
"corpus": [
  {
    "doc_id": "3048_BETRIEBSANLEITUNG_YUKONDAK_SI",
    "layer": "operational",
    "language_dominant": "sl",
    "pages": 158,
    "sha256": "aac6306152f579680cd4bcbeede70a1301e30ef75690fcf413881a02dee3109b"
  }
],
"bindings": [
  {
    "urn": "asset://3048/3048-CAD-159825-ZUSAMMENBAU/IMG-isometrics-balloons",
    "layer": "mechanical",
    "type": "assembly_position",
    "display_name": "isometrics_balloons",
    "caption": "Isometric assembly views with BOM position callouts (1-23)",
    "region_id": "3048_CAD_159825_ZUSAMMENBAU_p01_isometrics_balloons",
    "crop_file": "crops/3048_CAD_159825_ZUSAMMENBAU_p01_isometrics_balloons.png",
    "source_doc": "3048_CAD_159825_ZUSAMMENBAU",
    "source_page": 1
  }
]
}

```

Field rules

- **machine_id** is the corpus subfolder name; each machine has its own catalog file. The field is repeated inside the JSON for self-containment.
- **corpus** is a flat list — each entry lets the consumer verify it is reading the same source files the library was built from. **sha256** is the durable identity.
- **bindings** contains **only** assets with `HITL_approval: true`. The **needs_verification** flag is informational and does not gate export.

- `caption` is the short semantic label, stored inline rather than as a file reference.
- Unapproved assets are silently skipped — they stay in the inventory assets database for the human to triage. The export summary reports the skipped count.

Consumer guarantees

- Every binding's `source_doc` matches a `corpus[].doc_id`, and that document's `sha256` is the hash of the file the binding was derived from.
- The `bindings` list is the complete, authoritative set of citable URNs for this machine — any URN not present is either unreviewed or doesn't exist at all.

Layering

The rendering layer has no LLM dependency and no awareness of the application layer. The application layer imports the rendering layer; the reverse never happens.

```
corpus discovery + path resolution
    ▲
rendering primitives (grid overview, high-res crop, page info)
    ▲
shared agent toolkit (tool handlers, tool-use loop)
    ▲
cropping agents (the region-proposal agent and the re-cropping agent)
```

Keeping this arrow unidirectional ensures the rendering primitives stay deterministic and unit-testable, and the application layer can swap LLMs, prompts, or agent strategies without touching the rendering code.